Schulich School of Business

Music & Mental Health: Analyzing The Relationship Between Listening Habits & Self-Reported Mental Health

Final Project Documentation

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Problem Overview

This project explores the relationship between music and mental health and whether or not it improves, worsens or has no impact on mental health. Music is commonly used for therapeutic reasons, particularly to manage stress, reduce anxiety and improve mood. However, there is an increasing concern raised by many, that excessive use of music or intense genres may negatively impact mental health and contribute towards anxiety, depression, insomnia or OCD. Based on how frequently people, especially younger audiences, listen to music on a daily basis, it is important to understand this relationship. Music has become integrated in many individual's daily routines, whether it's for studying, working or coping, making its impact on mental health significant. The core problem stems from the lack of data-driven research making it difficult to draw reliable conclusions.

To address this gap, this project looks into a publicly available survey dataset to figure out patterns between individual music listening habits and mental health. Psychological well-being was evaluated across four different dimensions - anxiety, depression, insomnia, and OCD - using self-reported scores on a 0-10 scale. Additionally, variables relating to the impact of music on those dimensions were measured, including daily listening duration (hours per day), favorite genre, genre intensity classification (high vs. moderate vs. low), musical tempo (BPM), and per-genre listening frequency.

An important limitation of the study is its use of an entirely observational and self-reported dataset. No experiments were performed as part of the project, and the participants were not randomly assigned to any of the listening conditions. Thus, all of the findings of this project are based upon the observation that certain variables are correlated with others, but do not indicate any causal relationship between those variables.

<u>Research Questions</u>

This project was structured around four research questions:

1. Is there a relationship between the number of hours that individuals listen to music each day and their depression scores?
2. Are the mental health scores of individuals who listen to music excessively different from those who listen to music in moderation?
3. Are specific music genres associated with particular mental health outcomes such as elevated anxiety or insomnia?
4. Of the variables considered (hours listened, genre, BPM, or listening frequency), which is the best predictor of whether music either improves or worsens an individual's mood?

<u>Hypotheses</u>

Hypothesis 1 (Coping Mechanism): We hypothesized that the relationship between listening to music and depression would be nonlinear. Moderate levels of listening (2-4 hours per day) would be associated with lower scores for depression, while excessive listening (8 or more hours per day) would be associated with higher scores for depression - showing that individuals listened to music as a form of coping or isolation rather than as a therapeutic approach.

Hypothesis 2 (Genre Intensity): We hypothesized that individuals who have a preference for genres of high intensity (Metal, EDM, Rap, Hip Hop) will score higher on the insomnia and anxiety scales than those who prefer genres of low intensity (Classical, Jazz, Lofi).

Hypothesis 3 (BPM & Energy): We hypothesized a direct positive correlation between the BPM of the music that individuals prefer and their score for insomnia; we predicted that individuals who prefer music with higher BPM will have higher insomnia scores than those who prefer music with lower BPM.

Dataset Description

The dataset used in this project is the Music & Mental Health Survey Results dataset, available on Kaggle. This dataset contains 736 participants' responses to 33 different questions. Following the data cleaning steps described below, 720 rows of data remained, or 97.8% of the original dataset.

The survey captures information in two broad categories. The first category collects information regarding how many hours per day a respondent listens to music, their favourite genre out of 16 possible options, their self-reported preferred BPM, how frequently they listen to each genre on a four-level scale ranging from Never to Very Frequently, whether they listen while working, and whether they identify as an instrumentalist or composer. The second category collects information regarding the mental health of the individual; specifically, on a scale of 0 to 10, how severely they experience each of four conditions: anxiety, depression, insomnia, and OCD. These mental health scores are the primary variable of interest for the analysis of the dataset.

A separate variable, Music Effects, records whether the individual feels that music improves, worsens, or has no effect upon their mood. For the classification portion of this project, this variable was converted to a binary variable; variables with the value of Improve were assigned a value of 1, while variables with the value of Worsen or No Effect were assigned a value of 0. The distribution of these variables was approximately 75% Improve and 25% Worsen or No Effect, indicating a 3:1 class imbalance. Finally, the variables for genre frequency were converted from text values to numeric values; those who reported that they listen to music in a certain genre Never was assigned a value of 0, Rarely was assigned a value of 1, Sometimes was assigned a value of 2, and those who reported that they listen to music in a certain genre Very Frequently was assigned a value of 3.

<u>Data Cleaning Steps</u>

The following steps for cleaning the data were applied:

- Dropped the 'Timestamp' and 'Permissions' columns, which didn't contain any analytical value.
- BPM outliers were filtered: any values below 30 or above 300 were removed. The raw dataset contained at least one entry 1,000,000,000 BPM (which was a clear data entry error).
- Missing BPM values were imputed using each genre's median BPM as a replacement value rather than dropping those rows - this helped keep around 104 additional respondents.
- A single missing Age value was replaced as well using the dataset median.
- Rows with missing values for critical variables (Music effects, Hours per day, Anxiety, Depression, Insomnia, or Fav genre) were dropped (8 rows total)
- Any remaining missing strings values as well were filled with the mode (the mode value that is in the column).

Instructions To Run The Code

Requirements:

No local installation is needed. This project can run entirely on google colab, which is a free cloud-based Python environment with all the necessary libraries pre-installed (however, the only requirement is a google account). The following libraries are used and are all available natively in Colab without any additional installation: pandas, numpy, matplotlib, seaborn, scikit-learn, and scipy.

Steps To Run:

1. Open the Github repository and find the "Final_Code.ipynb"
2. Click the "Open in Colab" button at the top of the notebook file, or go to colab.research.google.com, select File > Open Notebook > Github, and paste the url.

3. Upload the dataset: in the Colab left sidebar, click the Files icon, then upload, and select mxmh_survey_results.csv from your local machine (remember, the file needs to be re-uploaded each session as Colab does not keep uploaded files between sessions)
4. Run all the cells by selecting Runtime > Run All from the top menu
5. Part 1 (the Data Preprocessing) will automatically generate cleaned_mxmh_data.csv in the colab session (note that this file will be lost when the Colab session ends unless you download it first via the Files sidebar).
6. All 12 visualizations will also show beneath their respective cells (Part 3 and 4 will show statistical results in the console output directly below each cell).

Design Decisions and Trade-Offs

1. BPM Imputation vs. Row Dropping

BPM had 107 missing values (14.5% of the dataset). Discarding those rows would have reduced the dataset to approximately 616 usable rows. Rather, choosing to impute the missing BPM values with the median BPM of each genre has the advantage that Metal fans are more likely to have a preferred BPM of around 140-160 rather than the median of approximately 120 BPM for all genres. The one potential trade-off of this approach is that Hypothesis 3 (regarding the relationship between BPM and insomnia) will be tested using the imputed values for BPM instead of the actual values.

2. Polynomial Regression for Hypothesis 1

Given the non-linear (inverted-U) relationship that was hypothesized between listening hours and depression for Hypothesis 1, fitting a linear regression model would have been inappropriate. Instead, a regression model that incorporated variables for the effect of listening hours as a quadratic variable (degree 2 polynomial) is what was done. Furthermore, the resulting coefficient of determination (R^2) was only approximately 0.022 for this model, which indicates that the variable of listening hours was only able to explain approximately 2.2% of the variance in depression scores. Thus, the model indicates that while the shape of the curve was in agreement with the hypothesis, the variable of daily listening hours was a weak predictor of depression scores - another finding that is worth considering in relation to the complexity of mental health conditions.

3. Handling Class Imbalance

The 3:1 class imbalance within the target variable indicates that 75% of classes are “Improve” whereas only 25% are “Worsens” or have no effect on the task. For these classes, any model that is created will achieve approximately 75% accuracy. While this is statistically high for model accuracy, it is analytically useless for evaluating the performance of the models. By applying the class_weight parameter with a value of “balanced” to the Logistic Regression, Decision Tree, and Random Forest models, each of these models will reduce its accuracy score but increase the Recall and F1 score for the “Worsens” and “No Effect” classes. Thus, the F1 score will be utilized as the main evaluation metric for all models within Part 4.

4. Stratified K-Fold Cross-Validation

Rather than one split of the data, we use stratified five-fold cross validation to ensure that each fold maintains the same distribution of the labels. Then, we report the average F1 score and the standard deviation of the F1 scores across the five folds, as well as the test set metrics.

5. Statistical Testing with One-Way ANOVA

The boxplots are visually suggestive of differences in the distributions of the variables across the groups. However, to confirm this visually suggestive finding statistically, a one-way analysis of variance (ANOVA) test can be performed. A one-way ANOVA tests whether the differences among the means of

the groups are greater than would be expected by chance. This test was applied to the variables of Anxiety and Insomnia across the three groups of music genres. The ANOVA result for Anxiety was statistically significant ($p = 0.0478$), indicating that respondents who prefer high-intensity genres show meaningfully higher anxiety scores on average than those who prefer low-intensity genres. The result of the test for Insomnia was not significant ($p = 0.3612$). Both results are reported in full rather than being omitted, including the non-significant finding for Insomnia.

Example Outputs

Figure 1. Polynomial Regression: Hours per Day vs. Depression Score

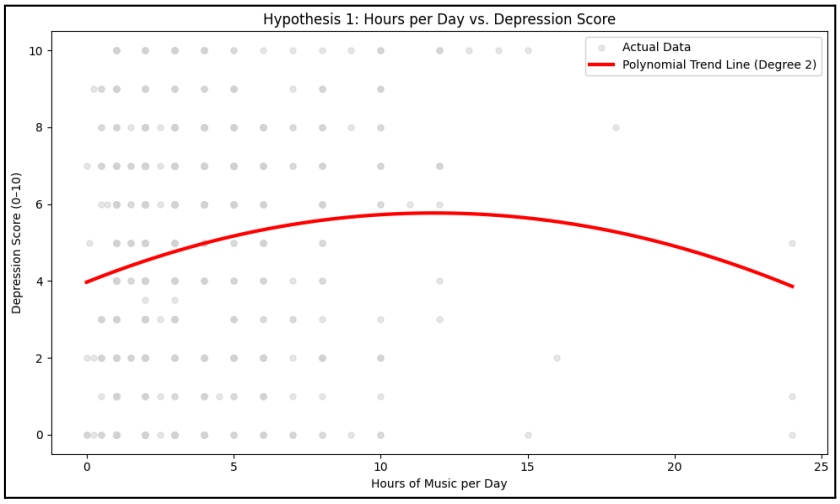

Figure 1 shows the polynomial regression fit for Hypothesis 1, looking at the relationship between daily listening hours and self-reported depression scores. The curve follows the expected inverted-U shape - depression scores go up as listening increases to a moderate level, then go down at higher levels. However, the R^2 value of 0.022 means that listening hours by themselves explain only 2.2% of the differences in depression scores; this means that even though the general shape matches the hypothesis, daily listening time is a weak predictor on its own, which fits with the idea that mental health is influenced by many factors.

Figure 2. Mean Mental Health Scores by Genre Intensity Level with ANOVA Results

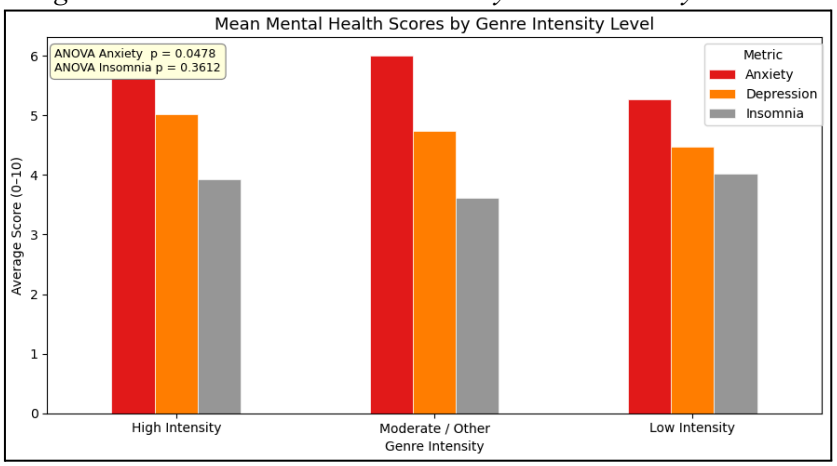

Figure 2 shows the average Anxiety, Depression, and Insomnia scores across three genre intensity groups (High: Metal, EDM, Rap, Hip Hop; Moderate/Other; Low: Classical, Jazz, Lofi), with one-way ANOVA p-values shown directly on the chart. The ANOVA result for Anxiety is $p = 0.0478$, which means there is a statistically significant difference in average anxiety scores between the groups, giving partial support to Hypothesis 2. The ANOVA result for Insomnia is $p = 0.3612$, which is not significant, meaning genre intensity is not linked to meaningful differences in insomnia levels in this sample.

Figure 3. Classification Model Comparison: All Four Performance Metrics

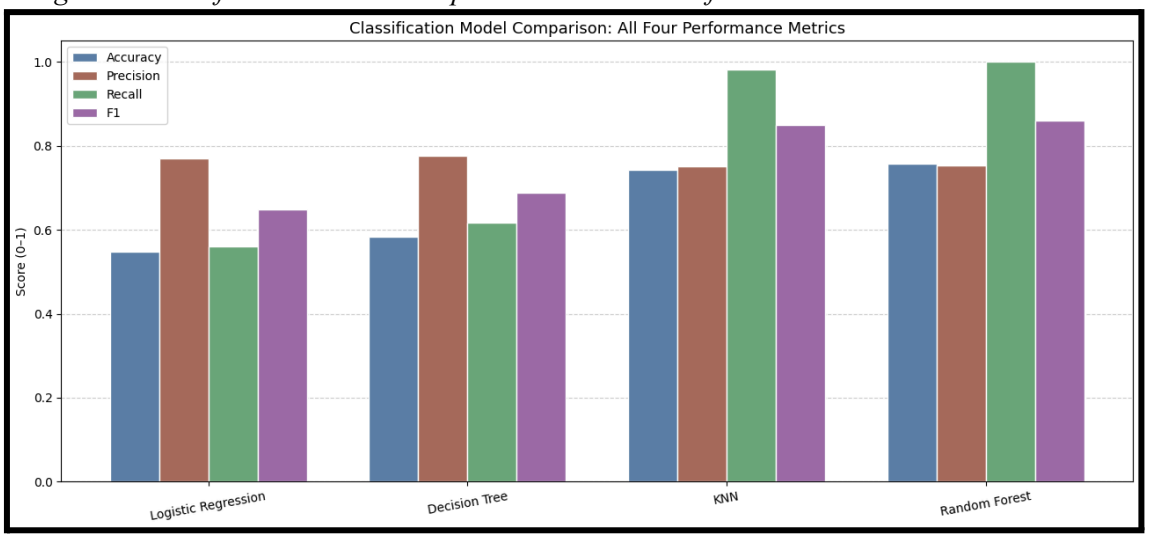

Figure 3 compares four models - Logistic Regression, Decision Tree, KNN, and Random Forest - using Accuracy, Precision, Recall, and F1-Score. KNN and Random Forest both reach F1 scores of about 0.85, clearly performing better than Logistic Regression (about 0.65) and Decision Tree (about 0.69). F1-Score is used as the main measure because the target variable has a 3:1 imbalance, where relying only on accuracy could give a misleading picture.

Figure 4. Random Forest Feature Importance Scores

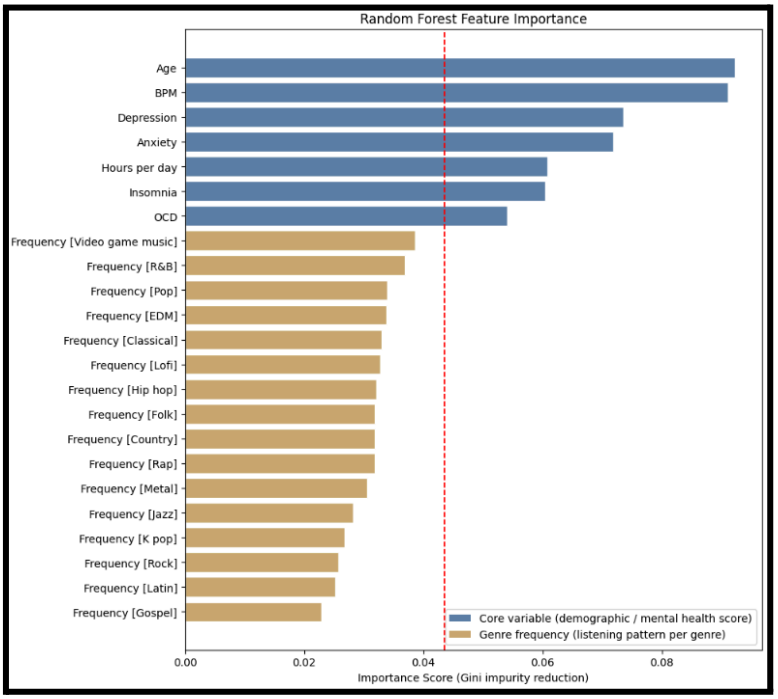

Figure 4 shows the Random Forest feature importance scores for all 23 input features, directly answering Research Question 4. Age is the strongest predictor of whether music improves or worsens mood (importance about 0.090), followed closely by BPM (about 0.089) and the main mental health scores - Depression and Anxiety. All 16 genre frequency features are below the average importance line (shown as a red dashed line), suggesting that listening to specific genres is a weak predictor on its own compared to demographic and mental health factors.

References

[1] Rasgaitis, C. (2022). Music & Mental Health Survey Results. Kaggle. https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results